John McGowan

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Education

Cornell University, College of Engineering — Ithaca, NY
B.S. in Computer Science — *May 2026*

- **Coursework:** Deep Learning, Machine Learning, Foundations of AI, NLP, Analysis of Algorithms, Data Structures & Functional Programming, Computer System Organization, Cryptography, Optimization, Financial Engineering, Linear Algebra
- **Awards:** Judges' Choice, Cornell Game Design Showcase; MGA Caddie Scholar; MLSAF Scholar

Experience

Meristead Wealth — June 2026 – Present
Trading Operations Analyst — *New York, NY*

- Built a centralized desktop application for the Trading Operations team's recurring morning tasks, backed by internal FastAPI services, saving the team 6 hours of manual reconciliation every morning across 15 to 20 daily tasks.
- Built and maintained integrations with the Salesforce, Orion, and Orion Eclipse REST APIs, replacing a weekly manual Orion-to-Salesforce upload with a daily automated Python sync that keeps CRM records within one day of Orion.
- Designed Azure-hosted infrastructure for internal automation services using Docker and PostgreSQL, giving the Trading Operations team secure, shared access to workflows and data through a single set of internal APIs.
- Built record-triggered Salesforce Flows on the firm's Client Information Sheet object, fixing false triggers by separating file uploads from Enhanced Notes so each flow fires only on real file attachments rather than on every saved note.
- Passed the SIE exam while supporting daily trading operations at a $1.3B AUM wealth management firm.

Mobius Emerging Opportunities Fund — Summer 2025
Financial Analyst Intern — *Dubai, UAE (Remote)*

- Built Python pipelines (pandas, NumPy) to ingest broker and custodian statements, reconcile positions and FX, and generate NAV and performance tables for weekly partner reviews, drawing on public market data and third-party vendor feeds.
- Developed a dashboard of daily portfolio positions, and reviewed portfolio and transaction data each day for discrepancies, working with investment and operations teams to resolve reporting issues before daily reports were distributed.

Projects

Ninth Circle: Social Album Log | *TypeScript, Next.js, SQLite, Docker* | ninthcircle.xyz — Aug 2025 – Present

- Built a full-stack Next.js/TypeScript app for rating albums with friends, with a social feed, search, profiles, and invite links, behind a provider interface for Spotify, Apple Music, and local data, pooling ratings for the same album across services.
- Implemented OAuth 2.0 + PKCE sign-in for Spotify and Apple with AES-256-GCM encrypted token storage, and blocked account takeover by refusing to auto-link on unverified provider emails; tested end to end against a local Spotify stub.
- Self-hosted on a Hetzner VPS: multi-stage Docker image running as non-root behind Caddy for automatic HTTPS, with a GitHub Actions pipeline that builds to GHCR and deploys over SSH, plus scripted server hardening and preflight checks.

Salesforce Data Loader Automation | *Python, Salesforce API, Linux* — 2026 – Present

- Replaced a paid dataloader.io subscription with a free in-house Python ETL tool that extracts Orion data, applies reconciliation and field-mapping rules, and upserts records daily through the Salesforce API, eliminating a recurring license fee.

Deep Learning Research | *Python, PyTorch, torchvision, scikit-learn* — Jan – May 2026

- Reproduced Brigato & Iocchi (2020), training CNNs of varying width and a deeper ResNet on CIFAR-10, Fashion-MNIST, and SVHN to study how model width and depth affect generalization from limited training data, using PyTorch and torchvision.

Ratatoskr, Squirrel Nightmares | *Java, LibGDX, Box2D* — Jan – May 2026

- As software engineering lead, led five developers and four UI/UX artists in the Cornell Game Design Initiative to build a Java desktop game on LibGDX with Box2D physics, slated for a Winter 2026 Steam release.

Offline Document Analysis Tool | *Python, FAISS, BGE embeddings* — Nov 2025

- Built an offline retrieval-augmented tool for querying private files: parsed and indexed 460 PDFs with BGE embeddings and FAISS semantic search to supply relevant context to a locally hosted 3B language model, keeping all data on the machine.

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Java, Apex, OCaml
Frameworks & Libraries: FastAPI, Next.js, React, Flask, pandas, NumPy, PyTorch, torchvision, scikit-learn, FAISS, LibGDX
Cloud & DevOps: Docker, Azure, Hetzner Cloud, GitHub Actions, Caddy, Git/GitHub
Databases & Platforms: PostgreSQL, SQLite, Linux, Windows, macOS
Integrations: Salesforce (Flows, REST API), Orion/Orion Eclipse APIs, Spotify API, MusicKit, dataloader.io, Power Automate

Activities

Alex Bores for Congress (NY-12) – Campaign Fellow — 2026
Cornell Brazilian Jiu Jitsu Club – Treasurer — Sept 2023 – May 2026
Pi Kappa Phi, Psi Chapter – Executive Board (Chaplain) — Jan 2022 – May 2026
Larchmont–Mamaroneck Youth Wrestling – Volunteer Coach — Sept 2018 – Present